

# Peripheral Oxygen Saturation (SpO<sub>2</sub>)

## What is SpO<sub>2</sub>?

During research on constructing a scatter plot for FiO<sub>2</sub> on Assignment 3, I encountered another vital sign that is closely related to FiO<sub>2</sub> and frequently used alongside it: **SpO<sub>2</sub>**. SpO<sub>2</sub> stands for **peripheral oxygen saturation** and **measures the percentage of oxygen carried in the blood** compared to the blood's maximum oxygen-carrying capacity. Oxygen is transported through the bloodstream to supply organs and tissues throughout the body. When oxygen levels become too low, organs may not receive sufficient oxygen to function properly, which can lead to serious complications (Mount Elizabeth, 2026).

SpO<sub>2</sub> readings are commonly **obtained using a pulse oximeter**. A pulse oximeter is a portable device that can be used to check blood oxygen levels by looking at how much light is absorbed by your blood.(Mount Elizabeth, 2026)

## What is its Clinical Range?

The World Health Organisation (2021) summarised below **how to interpret the readings** of the SpO<sub>2</sub> percentage.

1. **SpO<sub>2</sub> ≥ 94%** with no emergency signs (chest pain, dyspnoea, shortness of breath, altered mental status)- **normal reading**. Continue monitoring at home
2. **SpO<sub>2</sub> ≤ 94%**- patient **requires hospitalisation** for monitoring and further management.
3. **SpO<sub>2</sub> ≤ 90%** is a **medical emergency** and requires referral to a health facility with an intensive care unit or high dependency unit.

## What is its Importance to Triageing?

Oxygen is essential for human survival therefore, prolonged low oxygen levels in the bloodstream, known as hypoxemia, can negatively affect vital organs such as the brain, heart, liver, and kidneys. Hence,  $\text{SpO}_2$  is an extremely valuable vital sign in emergency triage systems.

An article examining emergency department overcrowding during the COVID-19 outbreak further demonstrated the importance of  $\text{SpO}_2$  when used together with  $\text{FiO}_2$  to evaluate patient severity. The study explained that the  $\text{SpO}_2/\text{FiO}_2$  ratio can be used to estimate the degree of hypoxemia (low oxygen levels) upon admission to the emergency department. According to the article, patients with an  $\text{SpO}_2/\text{FiO}_2$  ratio greater than 450, and without additional signs of respiratory distress, could potentially be managed as outpatients. In contrast, patients with an  $\text{SpO}_2/\text{FiO}_2$  ratio lower than 370 were more likely to require referral to intensive care units due to the increased severity of respiratory compromise (Catoire et al., 2021).

However, despite its importance,  $\text{SpO}_2$  should not be treated as a standalone variable that solely determines whether a patient requires immediate care. Like all other vital signs, it must be interpreted alongside additional vital signs. Nevertheless, incorporating  $\text{SpO}_2$  into triage assessments can provide healthcare workers with significant insight into a patient's respiratory condition and help prioritise patients who may require urgent medical attention.

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